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PLASMA PRECLEAN SYSTEM FOR CLUSTER TOOL

Abstract

A plasma processing system for cleaning a substrate is provided. The plasma processing system includes a process chamber that includes: a chamber body enclosing an interior volume; and a substrate support disposed in the interior volume. The plasma processing system includes a vacuum pump; a first exhaust line fluidly coupled between the interior volume of the process chamber and the vacuum pump; and a second exhaust line fluidly coupled between the interior volume of the process chamber and the vacuum pump. The first exhaust line and the second exhaust line are arranged to provide alternative paths for the exhaust between the interior volume and the vacuum pump, and the first exhaust line has an internal diameter that is at least 50% smaller than the internal diameter of the second exhaust line.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation application of co-pending U.S. patent application Ser. No. 17/836,657 filed Jun. 9, 2022, the entire contents of which is hereby incorporated herein by reference.

FIELD

[0002] Embodiments of the disclosure generally relate to the field of semiconductor manufacturing processes, and more particularly to a plasma preclean system and a cluster tool that includes the plasma preclean system.

BACKGROUND

[0003] A cluster tool is often used to perform processes (e.g., a deposition) on a substrate, such as a semiconductor substrate. For some depositions, such as epitaxial depositions, a plasma preclean process can be performed on substrate before the deposition. The plasma preclean process can be performed in one chamber of the cluster tool while the deposition can be performed in another chamber of the cluster tool. The cluster tools that include the deposition chamber and the plasma preclean chamber are generally configured to operate at vacuum pressures, such as pressures less than 10 Torr.

[0004] Recently, some epitaxial deposition processes operating at or near atmospheric pressure have been developed, but the plasma preclean process is still performed at a vacuum pressure. Having different portions of a cluster tool operate at substantially different pressures, such as a pressure of 10 Torr and a pressure of 760 Torr, can cause problems, such as the substrate being damaged, when the substrate is transferred between the environments having substantially different pressures.

[0005] Therefore, there is a need for methods and corresponding equipment to address the problems described above.

SUMMARY

[0006] In one embodiment, a plasma processing system for cleaning a substrate is provided. The plasma processing system includes a process chamber, comprising: a chamber body enclosing an interior volume; and a substrate support disposed in the interior volume. The plasma processing system includes a vacuum pump; a first exhaust line fluidly coupled between the interior volume of the process chamber and the vacuum pump; and a second exhaust line fluidly coupled between the interior volume of the process chamber and the vacuum pump, wherein the first exhaust line and the second exhaust line are arranged to provide alternative paths for the exhaust between the interior volume and the vacuum pump, and the first exhaust line has an internal diameter that is at least 50% smaller than the internal diameter of the second exhaust line.

[0007] In another embodiment, a plasma processing system for cleaning a substrate is provided. The plasma processing system includes a process chamber, comprising: a chamber body enclosing an interior volume; and a substrate support disposed in the interior volume. The plasma processing

system can further include a gas source fluidly coupled to the interior volume; a first supply line fluidly coupled between the interior volume and the gas source; and a second supply line fluidly coupled between the interior volume and the gas source, wherein the first supply line and the second supply line are arranged to provide alternative paths between the interior volume and the gas source, and the first supply line has an internal diameter that is at least 50% smaller than the internal diameter of the second supply line.

[0008] In another embodiment, a cluster tool for processing a semiconductor substrate is provided. The cluster tool includes a transfer chamber; a deposition chamber coupled to the transfer chamber; and a plasma processing system coupled to the transfer chamber, the plasma processing system comprising: a process chamber comprising: a chamber body enclosing an interior volume; and a substrate support disposed in the interior volume; a vacuum pump; a first exhaust line fluidly coupled between the interior volume of the process chamber and the vacuum pump; and a second exhaust line fluidly coupled between the interior volume of the process chamber and the vacuum pump, wherein the first exhaust line and the second exhaust line are arranged to provide alternative paths for the exhaust between the interior volume and the vacuum pump, and the first exhaust line has an internal diameter that is at least 50% smaller than the internal diameter of the second exhaust line.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] Implementations of the present disclosure, briefly summarized above and discussed in greater detail below, can be understood by reference to the illustrative implementations of the disclosure depicted in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical implementations of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective implementations.

[0010] FIG. 1 is a schematic top view of a cluster tool that can be used to perform the processes described herein, according to one embodiment.

[0011] FIG. 2 is a cross-sectional view of the preclean system from the cluster tool of FIG. 1, according to one embodiment.

[0012] FIG. 3 is a process flow diagram of a method of performing a preclean process on a substrate, according to one embodiment.

[0013] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. The figures are not drawn to scale and may be simplified for clarity. It is contemplated that elements and features of one implementation may be beneficially incorporated in other implementations without further recitation.

DETAILED DESCRIPTION

[0014] FIG. 1 is a schematic top view of a cluster tool **100** that can be used to perform the processes described herein, according to one embodiment. The cluster tool **100** includes a transfer chamber **102**, a pair of load lock chambers **106A**, **106B**, a preclean system **200** (also referred to as processing system), and a pair of process chambers **108**, **110**. The transfer chamber **102** includes a transfer robot **104** that can transfer a substrate **50** to and from each part of the cluster tool **100**.

[0015] As described in fuller detail below the preclean system **200** includes a number of features, which enable the preclean system **200** to perform a vacuum-pressure (e.g., 3 Torr), plasma preclean process on a substrate and then transfer the substrate to the transfer chamber **102** at atmospheric pressure or a pressure close to atmospheric pressure.

[0016] An overview of an exemplary process performed on a substrate **50** in the cluster tool is provided here. A substrate **50** can be received by the cluster tool **100** in one of the load lock

chambers **106A**, **106B**. Then the substrate **50** can be transferred by the transfer robot **104** through the transfer chamber **102** and into the chamber of the preclean system **200** for a preclean process, such as removal of oxides on the substrate **50**. Then, the substrate **50** can be transferred by the robot **104** through the transfer chamber **102** into one of the process chambers **108**, **110**, so that a process (e.g., an epitaxial deposition) can be performed on the substrate **50**. In some embodiments, a second process can be performed on the substrate **50** in the other process chamber **108**, **110**. After the processing in one or both of the process chambers **108**, **110**, the substrate **50** can then be transferred back by the transfer robot **104** through the transfer chamber **102** to one of the load lock chambers **106A**, **106B** for additional processing in other equipment (not shown).

[0017] The preclean system **200** can include a preclean chamber **201** (FIG. 2) configured to clean the substrate **50** prior to a process (e.g., an epitaxial deposition) to be performed on the substrate **50** in one of the process chambers **108**, **110**. In some embodiments, can include equipment for providing a plasma to the interior volume of the preclean chamber **201**, so that the preclean process can be performed on the substrate **50**. Depending on the embodiment, the preclean system can include one or more of a remote plasma source, a capacitively coupled plasma source, or an inductively coupled plasma source. The preclean system **200** can be used to perform the cleaning process described below.

[0018] In one embodiment, the process chambers **108**, **110** can be epitaxial deposition chambers. The process chambers **108**, **110** can be configured to perform epitaxial depositions at atmosphere pressure or pressures near atmosphere pressure (e.g., 700-800 Torr).

[0019] FIG. 2 is a cross-sectional view of the preclean system **200**, according to one embodiment. The preclean system **200** includes a preclean chamber **201** (also referred to as a process chamber). The preclean chamber **201** includes a chamber body **210**. The chamber body **210** includes a bottom **211**, a lid assembly **214**, and one or more chamber walls **212** connecting the bottom **211** with the lid assembly **214**. The chamber body **210** can enclose an interior volume **205** of the preclean chamber **201**.

[0020] The preclean chamber **201** further includes a substrate support assembly **216**. The substrate support assembly **216** can include a substrate support **232**, an actuator **234**, and a shaft **236** connecting the actuator **234** with the substrate support **232**. The substrate support **232** can be located in the interior volume **205** to support a substrate **50** during processing.

[0021] The chamber body **210** can further include a slit valve **215** to allow insertion and removal of a substrate **50** into and from the interior volume **205** of the preclean chamber **201**. The preclean system **200** and cluster tool **100** can be configured to have a pressure in the interior volume **205** remain below a pressure in the transfer chamber **102** when the slit valve **215** is opened to prevent flow of gas and/or particles from the preclean chamber **201** to the transfer chamber **102** as described in further detail below.

[0022] The lid assembly **214** is disposed at an upper end of the chamber body **210**. The lid assembly **214** can include a remote plasma source **220** for generating a plasma from cleaning gases provided to the remote plasma source **220**. The cleaning gases can be provided from a cleaning gas source **227** through a gas inlet **226** of the preclean chamber **201**. The cleaning gas source **227** can include a separate tank for each cleaning gas. In one embodiment, the cleaning gases from the cleaning gas source **227** can include one or more of hydrogen (H.sub.2), nitrogen trifluoride (NF.sub.3), and ammonia (NH.sub.3). The remote plasma source **220** can include a first electrode **221** and a second electrode **222**. The first electrode **221** can be spaced apart from the second electrode **222**. The remote plasma source **220** can include a plasma-generating volume **229** positioned between the first electrode **221** and the second electrode **222**.

[0023] The processing system **200** can include a radio frequency (RF) power source **224**. The RF power source **224** can be connected to the first electrode **221**. The second electrode **222** can be connected to electrical ground to serve as a return path for the RF power when the plasma is generated in the volume **229**. The RF power source **224** can be used to generate a plasma of the

cleaning gases inside plasma-generating volume **229** when the cleaning gases are provided to the remote plasma source **220**.

[0024] The lid assembly **214** can further include a blocker plate **228** and a showerhead **230** for distributing gas and/or plasma to the interior volume **205** of the preclean chamber **201**. The blocker plate **228** can be positioned between the remote plasma source **220** and the showerhead **230**. The blocker plate **228** can receive plasma and/or gas discharged from the remote plasma source **220**. In some embodiments, one or more gases may be provided directly to the blocker plate **228** or showerhead **230** allowing the remote plasma source **220** to be bypassed.

[0025] The processing system **200** can further include an inert gas source **240** connected to the preclean chamber **201**. In one embodiment, the inert gas source **240** includes nitrogen, but in other inert gases (e.g., argon) may also be used. The inert gas can be used to pressurize the interior volume **205** of the preclean chamber **201** after a preclean process is performed on the substrate **50** and/or before a new substrate **50** is transferred into the preclean chamber **201**. The processing system **200** can include a pressure sensor **260** configured to measure a pressure of the interior volume **205** of the process chamber **201**.

[0026] The inert gas source **240** can be connected to the gas inlet **226** of the process chamber through a first supply line **245** or a second supply line **246** of the processing system **200**. The first supply line **245** and the second supply line **246** can be connected to the gas inlet **226** through a common supply line **247**. The first supply line **245** and the second supply line **246** can be arranged to form parallel (i.e., alternative) paths relative to each other, so that gas can be supplied to the preclean chamber **201** through one of the supply lines without going through the other supply line.

[0027] The first supply line **245** can include a first supply valve **241** that can be opened to connect the first supply line **245** with the common supply line **247**. The second supply line **246** can include a second supply valve **242** that can be opened to connect the second supply line **246** with the common supply line **247**.

[0028] The first supply line **245** can have a smaller internal diameter relative to the internal diameter of the second supply line **246**. In some embodiments, the internal diameter of the first supply line **245** can be from about 5% to about 90%, such as from about 10% to about 50% of the internal diameter of the second supply line **246**. The smaller diameter of the first supply line **245** can be used to slowly raise the pressure in the interior volume **205** from the vacuum pressures (e.g., 2-20 Torr, such as between about 3-5 Torr) used for the preclean process after a preclean process is performed on the substrate **50**. On the other hand, the second supply line **246** can be used to quickly raise the pressure in the interior volume **205** back to atmospheric pressure or a pressure near atmospheric pressure after the pressure reaches a higher pressure (e.g., 300 Torr) from the gas provided from the smaller first supply line **245**. Slowly raising the pressure after the preclean process can prevent the likelihood of damaging the substrate **50** from an abrupt pressure change, such as mechanical damage caused by a wobbling or otherwise unintentionally moving the substrate **50**.

[0029] Using different supply lines with different internal diameters is one method of varying the rate at which gas is provided to the interior volume **205**. In other embodiments, the slower pressure changes can be achieved, for example, with an analog control valve on a single supply line. In some of these other embodiments, a sensor, such as a flowmeter or pressure sensor can be used to control the analog control valve or other actuator (e.g., a variable-speed pump) in order to control the rate at which the pressure in the interior volume **205** increases when the inert gas is supplied to the interior volume **205**, so that slower pressure changes in the interior volume **205** can be achieved.

[0030] The processing system **200** can further include a vacuum pump **218** configured to exhaust gas from the preclean chamber **201** through an exhaust port **223** of the preclean chamber **201**. The vacuum pump **218** can be connected to the exhaust port **223** through a first exhaust line **261** or a second exhaust line **262** of the processing system **200**. The first exhaust line **261** and the second

exhaust line **262** can be arranged to form parallel (i.e., alternative) paths relative to each other, so that gas can be exhausted from the preclean chamber **201** through one of the exhaust lines without going through the other exhaust line. The first exhaust line **261** and the second exhaust line **262** can be connected to the exhaust port **223** through a common exhaust line **263**. The first exhaust line **261** can include a first exhaust valve **219** that can be opened to fluidly couple the first exhaust line **261** with the common exhaust line **263**. The second exhaust line **262** can include a second exhaust valve **239** that can be opened to fluidly couple the second exhaust line **262** with the common exhaust line **263**.

[0031] The first exhaust line **261** can have a smaller internal diameter relative to the internal diameter of the second exhaust line **262**. All references provided in this disclosure to internal diameters also apply to internal cross-sectional areas, for example if the component (e.g., a fluid conduit) has a non-circular cross-section. In some embodiments, the internal diameter of the first exhaust line **261** can be from about 5% to about 75%, such as from about 10% to about 50% of the internal diameter of the second exhaust line **262**. The smaller diameter of the first exhaust line **261** can be used to smoothly and slowly lower the pressure in the interior volume **205** from atmospheric pressure or a pressure near atmospheric pressure (e.g., 700-800 Torr), to a lower pressure, such as from about 400-650 Torr, such as about 600 Torr. The pressure reduction can be performed, for example, after a substrate **50** is transferred into the preclean chamber **201** from the transfer chamber **102**, which is maintained at atmospheric pressure or a pressure near atmospheric pressure. On the other hand, the second exhaust line **262** can be used to quickly lower the pressure in the interior volume **205** down to a pressure near the pressure used for the preclean plasma process, such as a pressure less than 50 Torr, such as about 100 mTorr to about 20 Torr, such as a pressure between about 300 m Torr and about 5 Torr). Slowly lowering the pressure after a substrate **50** is transferred into the preclean chamber **201** can prevent the likelihood of damaging the substrate **50** from an abrupt pressure change, such as mechanical damage caused by wobbling or otherwise unintentionally moving the substrate **50**.

[0032] Using different exhaust lines with different internal diameters is one method of varying the rate at which gas and/or plasma is exhausted from the interior volume **205**, so that slower pressure changes in the interior volume **205** can be achieved. In other embodiments, the slower pressure changes can be achieved, for example, with an analog control valve on a single exhaust line. In some of these other embodiments, a sensor, such as a flowmeter or pressure sensor can be used to control the analog control valve or other actuator (e.g., a variable-speed vacuum pump) in order to control the rate at which the pressure in the interior volume **205** decreases when the interior volume **205** is brought down to a vacuum pressure for performing the plasma preclean process.

[0033] As introduced above, the substrate support assembly **216** includes the substrate support **232**, the actuator **234**, and the shaft **236** connecting the actuator **234** with the substrate support **232**. The shaft **236** can extend through a centrally-located opening formed in the bottom **211** of the chamber body **210**. The actuator **234** may be flexibly sealed to the bottom **211** of the chamber body **210** by bellows (not shown) that prevent vacuum leakage from around the shaft **236**. The actuator **234** allows the substrate support **232** to be moved vertically within the chamber body **210** between a process position and a lower transfer position. The transfer position can be slightly below the opening of the slit valve **215** formed through one of the one or more walls **212** of the chamber body **210**.

[0034] Although not shown, in some embodiments, an RF and/or DC bias can be coupled to the substrate support **232** to assist with directing the cleaning plasma toward the substrate **50**.

[0035] The processing system **200** can further include an auxiliary exhaust assembly **270**. The auxiliary exhaust assembly **270** can include a first auxiliary exhaust line **275**, a second auxiliary exhaust line **276**, and a common auxiliary exhaust line **278**. The auxiliary exhaust assembly **270** can further include a vacuum pump or other device for creating a negative pressure in the auxiliary exhaust assembly **270** lines relative to the interior volume **205** of preclean chamber **201**, so that gas

is exhausted from the interior volume **205** through the auxiliary exhaust assembly **270** when the valves of the auxiliary exhaust assembly **270** are opened.

[0036] The common auxiliary exhaust line **278** can be connected to the interior volume **205** of the preclean chamber **201**. The first auxiliary exhaust line **275** and the second auxiliary exhaust line **276** can be connected to the interior volume **205** of the preclean chamber **201** through the common auxiliary exhaust line **278**. The first auxiliary exhaust line **275** can include a first auxiliary exhaust valve **272** that can be opened to connect the first auxiliary exhaust line **275** with the common auxiliary exhaust line **278**. The second auxiliary exhaust line **276** can include a second auxiliary exhaust valve **274** that can be opened to connect the second auxiliary exhaust line **276** with the common auxiliary exhaust line **278**.

[0037] The first auxiliary exhaust valve **272** can be opened when a high pressure condition occurs. The first auxiliary exhaust line **275** can include a pressure sensor **271** to measure a pressure inside the first auxiliary exhaust line **275**. Upon measuring a pressure above a given threshold (e.g., 800 Torr), the first auxiliary exhaust valve **272** can be opened to relieve pressure inside the interior volume **205**. Because the preclean chamber **201** is operated at a higher pressure than other preclean chambers that typically operate at vacuum pressures (e.g., less than 100 Torr) for the preclean process and substrate transfer, more components of the preclean chamber are fastened or otherwise secured to each other. For example, the components of the lid assembly **214** can be secured to other components in the lid assembly **214** and/or to the chamber walls **212**. Some of these components in the lid assembly are generally unfastened for preclean chambers that operate at vacuum pressures for the preclean process and substrate transfer to and from the preclean chamber. The additional fastening of components in the preclean chamber **201** can help prevent movement of any of the components during the pressure changes that occur for each substrate preclean and transfer as described in further detail below. Securing these components though can create a safety issue as the previously unsecured components could move to relieve a high pressure situation. In the preclean chamber **201**, the first auxiliary exhaust valve **272** can open to relieve a high pressure condition when measured by the pressure sensor **271** and prevent an unsafe high-pressure condition from occurring.

[0038] The second auxiliary exhaust valve **274** can be opened when the slit valve **215** is opened, which allows gas to flow from the interior volume **205** and out the auxiliary exhaust assembly **270**. The interior volume **205** of the preclean chamber **201** is generally considered to be less clean than the interior volume of the transfer chamber **102**. Thus, gas should not flow from the interior volume **205** of the preclean chamber **201** to the interior volume of the transfer chamber **102**. Opening the second auxiliary exhaust valve **274** when the slit valve **215** opens reduces the pressure in the interior volume **205** relative to the pressure in the interior volume of the transfer chamber **102** and gas flows from the interior volume of the transfer chamber **102** through the interior volume **205** of the preclean chamber **201** and out through the auxiliary exhaust assembly **270**.

[0039] The processing system **200** can also include a controller **290** for controlling processes within the processing system **200** (FIG. 2) and other portions of the cluster tool **100** (FIG. 1). The controller **290** can be any type of controller used in an industrial setting, such as a programmable logic controller (PLC). The controller **290** includes a processor **292**, a memory **294**, and input/output (I/O) circuits **296**. The controller **290** can further include one or more of the following components (not shown), such as one or more power supplies, clocks, communication components (e.g., network interface card), and user interfaces typically found in controllers for semiconductor equipment.

[0040] The memory **294** can include non-transitory memory. The non-transitory memory can be used to store the programs and settings described below. The memory **294** can include one or more readily available types of memory, such as read only memory (ROM) (e.g., electrically erasable programmable read-only memory (EEPROM), flash memory, floppy disk, hard disk, or random access memory (RAM) (e.g., non-volatile random access memory (NVRAM)).

[0041] The processor **292** is configured to execute various programs stored in the memory **294**, such as a program configured to execute the method **1000** described below in reference to FIG. **3**. During execution of these programs, the controller **290** can communicate to I/O devices (e.g., sensors and actuators) through the I/O circuits **296**. For example, during execution of these programs and communication through the I/O circuits, the controller **290** can control outputs (e.g., open and close valves) and receive information from feedback devices (e.g., feedback on the open/close state of valves), sensors, and other instrumentation in the processing system **200** and other portions of the cluster tool **100**.

[0042] The memory **294** can further include various operational settings used to control the processing system **200** and other portions of the cluster tool **100**. For example, the settings can include pressure settings for when a transition between slowly changing and more quickly changing the pressure in the interior volume **205** is made in the method **1000** as described below in reference to FIG. **3** among various other settings.

[0043] FIG. **3** is a process flow diagram of a method **1000** of performing a preclean process on a substrate **50**, according to one embodiment. The method **1000** is described in reference to FIGS. **1-3**. The controller **290** can execute a program stored in memory to perform the method **1000**. The method begins at block **1002**.

[0044] At block **1002**, a new substrate **50** is transferred into the preclean chamber **201** from the transfer chamber **102** of the cluster tool **100**. At block **1002**, the interior volume of the transfer chamber **102** and the interior volume **205** of the preclean chamber **201** are both at atmospheric pressure or close to atmospheric pressure. The slit valve **215** of the preclean chamber **201** can be opened, and the transfer robot **104** can position the substrate **50** on the substrate support **232** in the interior volume **205** of the preclean chamber **201**. The second auxiliary exhaust valve **274** can be opened when the slit valve **215** is opened or slightly before the slit valve **215** is opened, which allows gas to flow from the interior of the transfer chamber **102** to the interior volume **205** of the preclean chamber **201** and out through the auxiliary exhaust assembly **270**, which is connected to a low pressure outlet (not shown). Having the gas flow from the transfer chamber **102** to the preclean chamber **201** helps maintain a clean environment in the transfer chamber **102** and the rest of the cluster tool outside of the preclean chamber **201**.

[0045] At block **1004**, the slit valve **215** is closed and a slow pump down of the preclean chamber **201** is performed. When the slow pump down begins the interior volume **205** of the preclean chamber **201** is at atmospheric pressure or pressure close to atmospheric pressure. The slow pump down of the interior volume **205** is performed by energizing the vacuum pump **218** and opening the first exhaust valve **219**. The slow pump down continues until a first target exhaust pressure in the interior volume **205** is reached as measured by the pressure sensor **260**. In some embodiments, the first target exhaust pressure can be from about 400 Torr to about 700 Torr, such as about 600 Torr. The first exhaust valve **219** is located on the first exhaust line **261**, which has a smaller internal diameter than the second exhaust line **262**. The smaller diameter of the first exhaust line **261** can be used to smoothly and slowly lower the pressure in the interior volume **205** until the first target exhaust pressure is reached. In some embodiments, the slow pump down can take from about 5 seconds to about 100 seconds, such as about 20 seconds. The slow pump down helps prevent damage to the substrate **50** that could be caused by movement (e.g., wobbling or other unintentional movement) of the substrate **50** when the substrate **50** is exposed to an abrupt pressure change, for example if a faster pump down of the interior volume **205** was performed without first performing the slow pump down at block **1004**.

[0046] At block **1006**, a fast pump down of the interior volume **205** of the preclean chamber **201** is performed from the first target exhaust pressure reached at block **1004** (e.g., 600 Torr) to a lower target exhaust pressure as measured by the pressure sensor **260** for the preclean plasma process. The lower target exhaust pressure can be from about 100 mTorr to about 20 Torr, such as between about 300 mTorr to about 5 Torr. The fast pump down of the interior volume **205** is performed by

opening the second exhaust valve **239** to connect the interior volume **205** of the vacuum pump **218** through the second exhaust line **262**. The first exhaust valve **219** can be closed at block **1006**. The second exhaust line **262** has a larger internal diameter (e.g., 100% larger) relative to the first exhaust line **261** enabling the pressure of the interior volume **205** to be reduced more rapidly relative to the pressure changes during the slow pump down of block **1004**. In some embodiments, the fast pump down can take from about 3 seconds to about 50 seconds, such as about 10 seconds. [0047] At block **1008**, a plasma preclean process is performed on the substrate **50** in the interior volume **205** of the preclean chamber **201**. The pre cleaning process can include providing one or more cleaning gases (e.g., NF.sub.3, NH₃, and H.sub.2) to the remote plasma source **220**, generating a plasma of the one or more cleaning gases in the remote plasma source **220**, and providing the plasma of the one or more cleaning gases to the interior volume **205** of the preclean chamber **201** through the showerhead **230** until the cleaning process at block **1008** is complete (e.g., after a designated period of time). In one embodiment, the plasma of the one or more cleaning gases can be used to remove oxides from the substrate **50**.

[0048] At block **1010**, after the cleaning process at block **1008** is complete, a slow fill of the interior volume **205** of the preclean chamber **201** with an inert gas (e.g., N.sub.2) is performed. The plasma preclean process performed at block **1008** is performed at a vacuum pressure (e.g., 3 Torr). Thus, the slow fill process at block **1010** begins at the vacuum pressure at which block **1008** ended. The slow fill of the interior volume **205** of the preclean chamber **201** is performed by opening the first supply valve **241** on the first supply line **245**, which connects the inert gas source **240** to gas inlet **226** of the preclean chamber **201** through the first supply line **245**.

[0049] The slow fill at block **1010** continues until a first target supply pressure in the interior volume **205** is reached as measured by the pressure sensor **260**. In some embodiments, the first target supply pressure can be from about 100 Torr to about 500 Torr, such as about 300 Torr. The first supply valve **241** is located on the first supply line **245**, which has a smaller internal diameter than the second supply line **246**. The smaller diameter of the first supply line **245** can be used to smoothly and slowly increase the pressure in the interior volume **205** until the first target supply pressure is reached. In some embodiments, the slow fill can take from about 5 seconds to about 100 seconds, such as about 20 seconds. The slow fill helps prevent damage to the substrate **50** that could be caused by movement (e.g., wobbling or other unintentional movement) of the substrate **50** when the substrate **50** is exposed to an abrupt pressure change, for example if a faster fill of the interior volume **205** was performed without first performing the slow fill at block **1010**.

[0050] At block **1012**, a fast fill of the interior volume **205** of the preclean chamber **201** is performed from the first target supply pressure reached at block **1010** (e.g., 200 Torr) to a higher second target supply pressure as measured by the pressure sensor **260**. The higher second target supply pressure can be atmospheric pressure or a pressure close to atmospheric pressure. The fast fill of the interior volume **205** is performed by opening the second supply valve **242**, which connects the inert gas source **240** to gas inlet **226** of the preclean chamber **201** through the second supply line **246**.

[0051] The first supply valve **241** can be closed at block **1012**. The second supply line **246** has a larger internal diameter (e.g., 100% larger) relative to the first supply line **245** enabling the pressure of the interior volume **205** to be increased more rapidly relative to the pressure changes during the slow fill at block **1010**. In some embodiments, the fast fill can take from about 3 seconds to about 50 seconds, such as about 10 seconds.

[0052] At block **1014**, with interior volume **205** at atmospheric pressure or pressure near atmospheric pressure, the substrate **50** can be removed from the preclean chamber **201**. The slit valve **215** of the preclean chamber **201** can be opened, and the transfer robot **104** can remove the substrate **50** from the substrate support **232**. The second auxiliary exhaust valve **274** can be opened when the slit valve **215** is opened or slightly before the slit valve **215** is opened, which allows gas to flow from the interior of the transfer chamber **102** to the interior volume **205** of the preclean

chamber **201** and out through the auxiliary exhaust assembly **270**, which is connected to a low pressure outlet (not shown). Having the gas flow from the transfer chamber **102** to the preclean chamber **201** helps maintain a clean environment in the transfer chamber **102** and other portions of the cluster tool **100** outside the preclean chamber **201**.

[0053] After being removed from the preclean chamber **201** the precleaned substrate **50** can be transferred into one of the process chambers **108**, **110**, so that an additional process can be performed on the substrate **50**, such as an epitaxial deposition performed at atmospheric pressure or a pressure close to atmospheric pressure.

[0054] At block **1016**, a determination is made regarding whether the preclean process should be performed on another substrate **50**. If another substrate **50** is to be precleaned, then blocks **1002** to **1014** can be repeated for the next substrate **50**. This repetition can continue until a target number of substrates **50** are precleaned using the method **1000**.

[0055] The equipment and processes described in this disclosure enable a vacuum pressure (e.g., 3 Torr), plasma preclean process to be performed on a substrate in a preclean chamber when the preclean chamber is connected to equipment operating at substantially higher pressures, such as atmospheric pressures. For example, the other portions of the cluster tool **100**, such as the transfer chamber **102**, the process chambers **108**, **110**, and the load lock chambers **106A**, **106B** can all operate at a substantially higher pressure, such as atmospheric pressure. By starting the pressure changes for the interior volume **205** of the preclean chamber **201** in a slow manner with the slow pump down at block **1004** and the slow fill at block **1010**, and then increasing rate of pressure change for the interior volume **205** with the fast pump down at block **1006** and the fast fill at block **1012**, numerous substrates **50** can be precleaned in a fast and efficient manner while the risk of damaging the substrates **50** (e.g., mechanical damage caused by wobbling or otherwise unintentionally moving the substrate) from abrupt pressure changes remains low.

[0056] While the foregoing is directed to implementations of the present disclosure, other and further implementations of the disclosure may be devised without departing from the basic scope thereof

Claims

1. A method of processing a substrate comprising: positioning a substrate on a substrate support in an interior volume of a chamber; reducing a pressure in the interior volume by exhausting the interior volume of the chamber through a first exhaust line without exhausting the interior volume through a second exhaust line, wherein the exhausting the interior volume through the first exhaust line is performed until a first exhaust pressure in the interior volume is reached when the substrate is on the substrate support, and the first exhaust pressure is greater than 400 Torr; exhausting the interior volume of the chamber through the second exhaust line without exhausting the interior volume through the first exhaust line to further reduce the pressure in the interior volume, wherein the exhausting the interior volume through the first exhaust line is performed until a second exhaust pressure in the interior volume is reached when the substrate is on the substrate support; providing gas to the interior volume through a first supply line without providing the gas through a second supply line to raise the pressure in the interior volume, wherein the providing the gas to the interior volume through the first supply line is performed until a first supply pressure in the interior volume is reached when the substrate is on the substrate support, and the first supply pressure is less than 300 Torr; and providing gas to the interior volume through the second supply line without providing the gas through the first supply line to further raise the pressure in the interior volume, wherein the providing the gas to the interior volume through the second supply line is performed until a second supply pressure in the interior volume is reached when the substrate is on the substrate support.

2. The method of claim 1, further comprising removing the substrate from the interior volume of

the chamber after the second supply pressure is reached.

3. The method of claim 1, wherein the second exhaust pressure is less than 50 Torr.

4. The method of claim 1, wherein the interior volume is at atmospheric pressure when the substrate is positioned on the substrate support.

5. The method of claim 1, wherein the first exhaust line has an internal diameter that is at least 50% smaller than the internal diameter of the second exhaust line.

6. The method of claim 1, wherein the first supply line has an internal diameter that is at least 50% smaller than an internal diameter of the second supply line.

7. The method of claim 1, wherein the pressure in the interior volume is reduced at a faster rate when gas is exhausted from the interior volume through the second exhaust line than when gas is exhausted from the interior volume through the first exhaust line.

8. The method of claim 1, wherein the pressure in the interior volume is increased at a faster rate when gas is provided to the interior volume through the second supply line than when gas is provided to the interior volume through the first supply line.

9. The method of claim 1, wherein the second supply pressure is greater than 600 Torr.

10. A method of processing a substrate comprising: performing a process on the substrate disposed on a substrate support in an interior volume of a chamber; providing gas to the interior volume through a first supply line without providing the gas through a second supply line to raise a pressure in the interior volume, wherein the providing the gas to the interior volume through the first supply line is performed until a first supply pressure in the interior volume is reached when the substrate is on the substrate support, and the first supply pressure is less than 300 Torr; and providing gas to the interior volume through the second supply line without providing the gas through the first supply line to further raise the pressure in the interior volume, wherein the providing the gas to the interior volume through the second supply line is performed until a second supply pressure in the interior volume is reached when the substrate is on the substrate support.

11. The method of claim 10, further comprising removing the substrate from the interior volume of the chamber after the second supply pressure is reached.

12. The method of claim 11, wherein the interior volume is at atmospheric pressure when the substrate is removed from the interior volume.

13. The method of claim 10, wherein the second supply pressure is greater than 600 Torr.

14. The method of claim 10, wherein the pressure in the interior volume is increased at a faster rate when gas is provided to the interior volume through the second supply line than when gas is provided to the interior volume through the first supply line.

15. The method of claim 10, wherein the first supply line has an internal diameter that is at least 50% smaller than an internal diameter of the second supply line.

16. A method of processing a substrate comprising: providing gas to an interior volume of a chamber through a first supply line without providing the gas through a second supply line to raise a pressure in the interior volume, wherein the providing the gas to the interior volume through the first supply line is performed until a first supply pressure in the interior volume is reached when the substrate is on a substrate support in the interior volume, and the first supply pressure is less than 300 Torr; and providing gas to the interior volume through the second supply line without providing the gas through the first supply line to further raise the pressure in the interior volume, wherein the providing the gas to the interior volume through the second supply line is performed until a second supply pressure in the interior volume is reached when the substrate is on the substrate support, and the pressure in the interior volume is increased at a faster rate when gas is provided to the interior volume through the second supply line than when gas is provided to the interior volume through the first supply line.

17. The method of claim 16, further comprising removing the substrate from the interior volume of the chamber after the second supply pressure is reached.

18. The method of claim 17, wherein the interior volume is at atmospheric pressure when the

substrate is removed from the interior volume.

19. The method of claim 16, wherein the first supply line has an internal diameter that is at least 50% smaller than an internal diameter of the second supply line.

20. The method of claim 16, wherein the second supply pressure is greater than 600 Torr.
